

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_no_replay.
- Best TSPLIB holdout gap: phase6_no_replay.
- Best synthetic holdout gap: phase6_no_replay.

Run Metadata

- run_name: run_20260519_040959_r
- started_at_local: 2026-05-19 04:09:59
- finished_at_local: 2026-05-19 04:30:35
- duration_hhmm: 00:21
- duration_seconds: 1236.776
- seed_offset: 17000
- replicate_label: r
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.078538	0.0	0.049979	0.236772	0.1501	0.0	0.799532	0.76	0.025584
phase6_random_replay	random	score_only	False	0.104797	0.003941	0.068122	0.236772	0.175118	0.19457	0.850208	0.76	0.113967
phase6_failure_replay	failure	score_only	False	0.099242	0.0	0.063154	0.171206	0.300914	0.261637	0.786883	0.76	0.034508

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.195726	0.010343	0.128314	0.236772	0.163361	0.19457	0.0	0.76	0.0
phase6_stratified_random_replay	stratified_random	score_only	False	0.131903	0.0	0.083938	0.236772	0.159309	0.221162	0.829235	0.76	0.043764
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.133808	0.0	0.085151	0.236772	0.163657	0.157984	0.0	0.76	0.0
phase6_residual_failure_replay	residual_failure	score_only	False	0.213387	0.064916	0.159398	0.149274	0.338865	0.291666	0.851031	0.56	-0.029265
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.101901	0.0	0.064846	0.171206	0.269935	0.171012	0.78994	0.76	0.037351
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.175893	0.0	0.111932	0.171206	0.293959	0.172748	0.723666	0.76	0.02042

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.078538, synthetic 0.0, combined 0.049979.
- Accepted-epoch count 4, mean accepted code novelty 0.799532, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.1501, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency 0.025584 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.078538 across 7 instances; family means: ch=0.026933, kroD=0.03602, pcb=0.219642, pr=0.114295, rd=0.100759, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3104, "expected_gap": 0.400776, "family": "a", "name": "a280", "optimality_gap": 0.203567, "residual_gap": -0.197209}, {"best_known_cost": 629, "cost": 691, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.098569, "residual_gap": -0.158983},

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{"best_known_cost": 7542, "cost": 8121, "expected_gap": 0.245107, "family": "berlin", "name": "berlin52",  
"optimality_gap": 0.07677, "residual_gap": -0.168337}}
```

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.104797, synthetic 0.003941, combined 0.068122.
- Accepted-epoch count 2, mean accepted code novelty 0.850208, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.175118, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133843, mean selected residual gap 0.016398.
- Adaptation efficiency 0.113967 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.104797 across 7 instances; family means: ch=0.078327, kroD=0.154363, pcb=0.187365, pr=0.053782, rd=0.076233, st=0.105185.
- Panel synthetic_holdout mean gap 0.003941 across 4 instances; family means: clustered_gaussian=0.015764, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3376, "expected_gap": 0.400776, "family": "a", "name": "a280", "optimality_gap": 0.309035, "residual_gap": -0.091741}, {"best_known_cost": 7542, "cost": 9473, "expected_gap": 0.245028, "family": "berlin", "name": "berlin52", "optimality_gap": 0.256033, "residual_gap": 0.011005}, {"best_known_cost": 14379, "cost": 17487, "expected_gap": 0.326977, "family": "lin", "name": "lin105", "optimality_gap": 0.216149, "residual_gap": -0.110828}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.099242, synthetic 0.0, combined 0.063154.
- Accepted-epoch count 4, mean accepted code novelty 0.786883, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.300914, size bias 0.120322, and failure concentration 0.261637.
- Replay selection diversity 0.21241, mean selected expected gap 0.304752, mean selected residual gap 0.011468.
- Adaptation efficiency 0.034508 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 8, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 4.
- Panel heldout_tsplib mean gap 0.099242 across 7 instances; family means: ch=0.072119, kroD=0.063492, pcb=0.229962, pr=0.099677, rd=0.096587, st=0.060741.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3458, "expected_gap": 0.401939, "family": "a", "name": "a280", "optimality_gap": 0.34083, "residual_gap": -0.061109}, {"best_known_cost": 14379, "cost": 16397, "expected_gap": 0.32496, "family": "lin", "name": "lin105", "optimality_gap": 0.140344, "residual_gap": -0.184616}, {"best_known_cost": 7542, "cost": 8194, "expected_gap": 0.240918, "family": "berlin", "name": "berlin52", "optimality_gap": 0.086449, "residual_gap": -0.154469}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.195726, synthetic 0.010343, combined 0.128314.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.163361, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132156, mean selected residual gap -0.000676.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.195726 across 7 instances; family means: ch=0.259572, kroD=0.250305, pcb=0.241778, pr=0.176019, rd=0.157649, st=0.025185.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3452, "expected_gap": 0.403645, "family": "a", "name": "a280", "optimality_gap": 0.338503, "residual_gap": -0.065142}, {"best_known_cost": 14379, "cost": 19077, "expected_gap": 0.321761, "family": "lin", "name": "lin105", "optimality_gap": 0.326726, "residual_gap": 0.004965}, {"best_known_cost": 629, "cost": 755, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.200318, "residual_gap": -0.0531}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.131903, synthetic 0.0, combined 0.083938.
- Accepted-epoch count 2, mean accepted code novelty 0.829235, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.159309, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.236082, mean selected residual gap -0.026283.
- Adaptation efficiency 0.043764 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 8, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.131903 across 7 instances; family means: ch=0.134471, kroD=0.224289, pcb=0.173618, pr=0.096349, rd=0.115676, st=0.044444.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3421, "expected_gap": 0.406514, "family": "a", "name": "a280", "optimality_gap": 0.326483, "residual_gap": -0.080031}, {"best_known_cost": 14379, "cost": 18147, "expected_gap": 0.321274, "family": "lin", "name": "lin105", "optimality_gap": 0.262049, "residual_gap": -0.059225}, {"best_known_cost": 629, "cost": 754, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.198728, "residual_gap": -0.057552}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.133808, synthetic 0.0, combined 0.085151.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.163657, size bias 8.3e-05, and failure concentration 0.157984.
- Replay selection diversity 0.26167, mean selected expected gap 0.167861, mean selected residual gap -0.016493.

- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.133808 across 7 instances; family means: ch=0.096662, kroD=0.125998, pcb=0.205561, pr=0.216931, rd=0.169659, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19215, "expected_gap": 0.323194, "family": "lin", "name": "lin105", "optimality_gap": 0.336324, "residual_gap": 0.01313}, {"best_known_cost": 629, "cost": 810, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.287758, "residual_gap": 0.033704}, {"best_known_cost": 2579, "cost": 3232, "expected_gap": 0.406049, "family": "a", "name": "a280", "optimality_gap": 0.253199, "residual_gap": -0.15285}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.851031, and final complexity 0.56.
- Active replay archive residual_archive with mean diversity 0.149274, mean hardness 0.338865, size bias 0.093558, and failure concentration 0.291666.
- Replay selection diversity 0.177489, mean selected expected gap 0.264849, mean selected residual gap 0.087544.
- Adaptation efficiency -0.029265 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.404963, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.003645}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.129094}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.322373, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.00644}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.101901, synthetic 0.0, combined 0.064846.
- Accepted-epoch count 3, mean accepted code novelty 0.78994, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.269935, size bias 0.120322, and failure concentration 0.171012.
- Replay selection diversity 0.195273, mean selected expected gap 0.301938, mean selected residual gap -0.025534.
- Adaptation efficiency 0.037351 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.101901 across 7 instances; family means: ch=0.114136, kroD=0.060862, pcb=0.249813, pr=0.075001, rd=0.062326, st=0.037037.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3470, "expected_gap": 0.402482, "family": "a", "name": "a280", "optimality_gap": 0.345483, "residual_gap": -0.056999}, {"best_known_cost": 14379, "cost": 19140, "expected_gap": 0.321803, "family": "lin", "name": "lin105", "optimality_gap": 0.331108, "residual_gap": 0.009305}, {"best_known_cost": 7542, "cost": 8516, "expected_gap": 0.233068, "family": "berlin", "name": "berlin52", "optimality_gap": 0.129143, "residual_gap": -0.103925}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.175893, synthetic 0.0, combined 0.111932.
- Accepted-epoch count 2, mean accepted code novelty 0.723666, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.293959, size bias 0.120322, and failure concentration 0.172748.
- Replay selection diversity 0.194842, mean selected expected gap 0.298989, mean selected residual gap 0.000536.
- Adaptation efficiency 0.02042 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.175893 across 7 instances; family means: ch=0.124171, kroD=0.223537, pcb=0.270491, pr=0.216866, rd=0.19646, st=0.075556.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3469, "expected_gap": 0.401086, "family": "a", "name": "a280", "optimality_gap": 0.345095, "residual_gap": -0.055991}, {"best_known_cost": 7542, "cost": 9237, "expected_gap": 0.235826, "family": "berlin", "name": "berlin52", "optimality_gap": 0.224741, "residual_gap": -0.011085}, {"best_known_cost": 629, "cost": 743, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.18124, "residual_gap": -0.082671}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Results

Overview

- ****Best overall condition**** (held-out TSPLIB, synthetic holdout, and transfer):

phase6_no_replay

- Mean TSPLIB gap: 7.85%
- Mean transfer gap: 5.0%
- Mean synthetic gap: 0%
- No replay used (replay_mode: none)
- Archive size: 28 with moderate diversity (0.24) and medium hardness (0.17)
- Code novelty high (~0.80 mean)
- Nine total experimental conditions varying replay modes and selection modes.

Transfer Performance (Hold-out TSPLIB & Synthetic Holdout gaps)

Condition	TSP LIB Gap	Trans fer Gap	Synth etic Gap	Replay Mode	Code Novelty (mean)	Notes
phase6_no_replay	0.0785	0.0500	0.0000	none	0.799	Best transfer + optimality
phase6_random_replay	0.1048	0.0681	0.0039	random	0.850	Worse than no replay in gaps
phase6_failure_replay	0.0992	0.0632	0.0000	failure	0.787	Slightly better than random replay
phase6_random_replay_compression	0.1957	0.1283	0.0103	random (compression aware)	0.000	Significant drop in code novelty, worse gaps
phase6_stratified_random_replay	0.1319	0.0839	0.0000	stratified_random	0.829	Moderate performance
phase6_diversity_weighted_replay	0.1338	0.0852	0.0000	diversity_weighted	0.000	Code novelty dropped, worse gaps than no replay
phase6_residual_failure_replay	0.2134	0.1594	0.0649	residual_failure	0.851	Worst gaps, failure concentration highest
phase6_diversity_failure_replay	0.1019	0.0648	0.0000	diversity_failure	0.790	Comparable to failure replay
phase6_diversity_failure_replay_compression	0.1759	0.1119	0.0000	diversity_failure (compression)	0.724	Moderate degradation compared to non-compression

Key points on transfer gaps:

- ****No replay**** is best on both synthetic and TSPLIB holdouts and transfer gaps.
- Replay conditions generally worsen transfer and TSPLIB gaps.
- Compression-aware replay conditions (random and diversity_failure) lead to poorer code novelty and worse transfer gaps.
- Residual_failure replay shows poorest transfer gaps and highest failure concentration.

Replay Archive Analysis (Diversity, Hardness, Size Bias, Failure Concentration)

Condition	Archive Diversity	Archive Hardness	Archive Size Bias	Final Replay Failure Concentration	Mean Replay Failure Concentration	Mean Replay Selection Diversity
phase6_no_replay	0.240	0.174	0.0	0.0	0.0	0.0
phase6_random_replay	0.240	0.185	0.0	0.16	0.195	0.262
phase6_failure_replay	0.175	0.310	0.125	0.31	0.26	0.212

Condition	Archive Diversity	Archive Hardness	Archive Size Bias	Final Replay Failure Concentration	Mean Replay Failure Concentration	Mean Replay Selection Diversity
phase6_random_replay_compression	0.240	0.186	0.0	0.16	0.195	0.262
phase6_stratified_random_replay	0.240	0.187	0.0	0.25	0.22	0.187
phase6_diversity_weighted_replay	0.240	0.192	0.0	0.10	0.16	0.26
phase6_residual_failure_replay	0.150	0.339	0.094	0.33	0.29	0.177
phase6_diversity_failure_replay	0.175	0.309	0.125	0.33	0.17	0.20
phase6_diversity_failure_replay_compression	0.175	0.315	0.125	0.17	0.17	0.19

Observations:

- Failure replay modes (failure, residual_failure, diversity_failure) result in:
- Higher archive hardness (0.31-0.34) vs no replay (~0.17-0.19)
- Larger size bias (~0.12) vs none.
- Higher failure concentration (final up to 0.33) indicating replay focuses on difficult failures.
- Random and stratified replay maintain archive diversity near baseline (0.24).
- Residual failure replay has the lowest archive diversity (0.15) and highest hardness.
- Compression pressure in replay archives leads to code novelty reduction.

Code Novelty and Complexity

Condition	Mean Code Novelty	Last Code Novelty	Mean Complexity	Notes
phase6_no_replay	0.80	0.73	0.76	Highest code novelty, good complexity
phase6_random_replay	0.85	0.85	0.76	High novelty, no code novelty drop
phase6_failure_replay	0.79	0.79	0.76	Moderate novelty
phase6_random_replay_compression	0.00	0.00	0.76	Significant novelty loss due to compression
phase6_stratified_random_replay	0.83	0.83	0.76	High novelty, no complexity penalty
phase6_diversity_weighted_replay	0.00	0.00	0.76	Zero novelty, no improvement in transfer
phase6_residual_failure_replay	0.85	0.86	0.56	High novelty, lower complexity
phase6_diversity_failure_replay	0.79	0.80	0.76	Moderate novelty
phase6_diversity_failure_replay_compression	0.72	0.72	0.76	Drop in novelty with compression

Mechanism Analysis for Replay Modes

- **phase6_no_replay** (no replay) yields best transfer and held-out performance, maintaining high code novelty and moderate archive diversity/hardness.
- **Random replay** (phase6_random_replay and compression version):
 - Increases final archive hardness but leads to increased gaps on held-out and transfer datasets relative to no replay.
 - Compression significantly reduces code novelty and increases gaps, indicating a trade-off.
- **Failure replay** (phase6_failure_replay):
 - Produces archives with much higher hardness and some size bias.
 - Failure concentration is moderate-high (~0.3), indicating focus on failure cases.
 - Transfer and holdout gaps notably worse than no replay.
 - Code novelty somewhat lower but not absent.
- **Residual failure replay**:
 - Highest archive hardness and failure concentration.
 - Worst transfer and held-out gaps.
 - Code novelty remains high.
 - Complexity lower than others, possibly indicating overfitting or less diverse behaviors.
- **Diversity-weighted replay**:
 - Maintains archive diversity but results in zero code novelty and no transfer improvement.
 - Holds worse gaps on hold-out datasets.
 - Indicates lexical novelty without effective algorithmic transfer.
- **Diversity failure and its compression variant**:
 - Balanced archive diversity and hardness with moderate failure concentration.
 - Compression lowers code novelty and worsens transfer performance.
 - Non-compression version performs better than random or failure replay but worse than no replay.
- **Stratified random replay**:
 - Moderate improvements in replay selection diversity.
 - Transfer and held-out gaps better than random replay but still worse than no replay.
 - Code novelty remains high.

Conclusions

- The **best transfer and optimality** are achieved by the no replay condition (phase6_no_replay).
- All replay methods increase the difficulty (hardness) and failure concentration of the training sets but lead to worse generalization (higher transfer and heldout gaps).
- Use of compression in archive management reduces code novelty and degrades transfer performance, especially in random and diversity failure replay.
- High **code novelty** does not guarantee better transfer; for example, failure and residual failure replay show high novelty but poor transfer.
- Lexical/code novelty decline with replay compression without corresponding transfer gain suggests lexical novelty measures may mislead regarding genuine algorithmic improvement.
- **Replay modes focusing on failures or residual failures create archives with size bias and increased hardness but not better transfer**.
- Diversity-weighted replay achieves archive diversity but fails to improve transfer or preserve code novelty.
- Replay selection diversity is higher in random and diversity-weighted replay but does not correlate with better transfer performance.

Final Notes

- Prioritize phase6_no_replay for transfer and solution quality.
- Replay strategies induce more challenging but less generalizable training, increasing failure concentration and hardness.
- Compression reduces code novelty significantly, harming transfer.
- Lexical or archive diversity does not equate to algorithmic invention or improved generalization unless matched by transfer metrics.